SunTax CH

Next-Generation AI Swiss Tax Platform • System Architecture & Client Demo Handover

Client Evaluation Dossier
Date: October 2026
● Production Deployed

1. Live Cloud Endpoints

Component / Service	Live URL	Technical Details & Description
Frontend Web Application Vercel Production	**https://frontend-ruby-gamma-46.vercel.app/**	Next.js 14 App Router, TypeScript, responsive Swiss UI, live WebSocket updates.
Backend API Engine Render Cloud Service	**https://suntax-api.onrender.com**	FastAPI core with deterministic Swiss tax calculator, PyMuPDF OCR, PyOTP.
Interactive Swagger UI OpenAPI Documentation	**https://suntax-api.onrender.com/api/docs**	Interactive API testing sandbox for all authentication, document, and tax endpoints.
Technical Specification ReDoc Specification	**https://suntax-api.onrender.com/api/redoc**	Comprehensive OpenAPI 3.0 schema definitions, payloads, and error codes.
System Health Check	**https://suntax-api.onrender.com/health**	`Returns {'status': 'ok', 'environment': 'production'}.`

NOTE — Cloud Performance on Render Free Tier: The backend operates on Render free instance. If the service has been idle, the initial request awakens the container in ~45-60 seconds. All subsequent operations execute at sub-150ms latency.

2. GitHub Repository & Codebase Structure

Repository Detail	Configuration & URL	Architecture Notes
GitHub Repository	**https://github.com/usamashaikh13/suntax**	Monorepo with dedicated frontend/, backend/, and tax-rules/.
Production Branch	`main (Synced & Fully Tested)`	Continuous deployment triggered automatically on push to main.
Local Setup Helper	`./backend/start_dev.sh`	Automates venv creation, dependency installation, and server startup.

3. Database Architecture & Database File

Database Layer	File / Connection Target	Description & Security Policy
Development DB File	`backend/suntax_dev.db`	SQLite ACID database containing initialized user tables, tax returns, and profiles.
Production DB Engine	`PostgreSQL 16 (asyncpg)`	Asynchronous connection pool with Alembic schema migration management.
Data Isolation	`Row Level Security (RLS)`	User data strictly isolated; unauthorized cross-tenant queries blocked at DB layer.
Local Object Vault	`backend/dev_storage/`	Local encrypted file vault mirroring S3/MinIO bucket layout with user-prefixed keys.

4. Pre-Configured Demo Accounts

Account	Email & Password	2FA Security Status	Authenticator Setup / Code Access
Primary Demo (Full 2FA Active)	**client@suntax.ch** TestPassword123!	● **2FA Active** Required at Login & Final Submission	**Key:** OWZDANSQXHFBZXANGHT3ZKUYM2XW2AAV Or CLI: .venv/bin/python get_2fa.py client@suntax.ch
Secondary Demo (Standard Access)	**demo@suntax.ch** TestPassword123!	■ **Direct Login** Ready for live 2FA setup demo	Can be used to demonstrate live 2FA QR code pairing in *My Account* → *Security*.

5. Step-by-Step Client Testing Guide (5-Minute Verification)

Step 1: Authenticate	Navigate to /login on Vercel or localhost. Log in with **client@suntax.ch**. Enter the 6-digit TOTP code from your mobile authenticator app (Google Authenticator / Apple Passwords).
Step 2: Create Declaration	Click **'+ New Tax Return'**. Select Canton **ZH (Zürich)**, Municipality **Zürich (BFS #261)**, and Tax Year **2025**. Click Confirm.
Step 3: Upload Documents	In the **Documents** tab, drag & drop the test documents (Salary Certificate, UBS Bank Statement, Pillar 3a Certificate, Donation Receipt).
Step 4: Review OCR Data	Click **'Review Fields'** on any document marked Needs Review. Review the extracted figures and click **'Apply to Tax Profile'**.
Step 5: Compute Taxes	Switch to the **Calculation** tab and click **'Calculate Tax'**. Deterministic calculation resolves: Federal CHF 637, Canton CHF 1'852.40, Municipal CHF 2'204.35 (Total: CHF 4'693.75).
Step 6: Step-Up 2FA Submit	Under **Review & Submit**, check the Art. 110 DBG declaration box. Click **'Confirm Tax Return Draft'**. Enter your 6-digit 2FA security code to seal and lock the filing.

6. Key API Endpoints Reference

Method & Path	Endpoint Category	Operational Description
`POST /api/v1/auth/login`	Authentication	Validates email & password. If 2FA enabled, issues temporary token for TOTP challenge.
`POST /api/v1/auth/login/2fa`	2FA Challenge	Exchanges temporary token + 6-digit TOTP code for full-scope JWT access token.
`GET /api/v1/tax-returns`	Tax Filings	Retrieves paginated list of tax declarations owned by authenticated user.
`POST /api/v1/tax-returns/{id}/calculate`	Deterministic Engine	Applies verified cantonal and federal tax brackets to user's TaxProfile.
`POST /api/v1/tax-returns/{id}/confirm`	Legal Confirmation	Seals declaration under Art. 110 DBG; validates 6-digit 2FA code if TOTP enabled.
`POST /api/v1/tax-returns/{id}/export/pdf`	Official Export	Generates complete official Swiss tax declaration summary PDF.

7. Best Platform Recommendations: Storage & Deployment

For Swiss tax, financial, and wealth advisory software, infrastructure selection is heavily dictated by **data sovereignty**, **compliance with the Swiss Federal Act on Data Protection (FADP / nDSG)**, and **FINMA Circular 2018/3 (Outsourcing)**. Below is the expert evaluation of optimal hosting and storage tiers for SunTax production rollout.

A. Document Vault & Object Storage Recommendations

Platform Provider	Swiss Data Residency	Security & Compliance Features	Recommendation Level
Infomaniak Swiss S3 Geneva & Winterthur, CH	**100% Swiss Sovereign** Zero foreign CLOUD Act exposure	Native S3 REST API, AES-256 server-side encryption, ISO 27001, immutable object locks, Tier III+ data centers. Pricing: ~CHF 0.01/GB.	**Top Recommendation** (Best Swiss Compliance & Value)
Exoscale Object Storage Zürich & Geneva, CH	**100% Swiss Sovereign** Regulated Swiss entity	FINMA-ready, ISO 27001 / 27018 compliant, multi-zone Zurich replication, high throughput for financial institutions.	**Top Enterprise Choice** (Swiss Banking Grade)
AWS S3 (Zurich Region) Region: eu-central-2	**Swiss Data Center** (Subject to US CLOUD Act)	AWS KMS with customer-managed keys (CMK), S3 Glacier Instant Retrieval for historical declarations, 99.999999999% durability.	**Enterprise Scaler** (If corporate AWS in place)
Self-Hosted MinIO On-Prem / Private VPS	**Customer Controlled** (Bare-metal / Private Cloud)	Open-source S3 compatible, full control over hardware keys and network perimeters. Requires internal maintenance.	**Private Sovereign** (On-premise setups)

B. Application & Database Deployment Recommendations

Platform & Role	Architecture Pattern	Operational Benefits & Scalability	Best Suited For
Vercel Enterprise / Pro Frontend Web Layer	Edge SSR (Frankfurt/Zurich)	Instant atomic deployments, global AnyCast CDN, automatic HTTPS, sub-50ms TTFB for Swiss taxpayers.	**Recommended Frontend** (Zero devops overhead)
Exoscale SKS / Cloud VPS Backend & Celery Workers	Managed K8s / Linux in Zürich	Pure Swiss hosting, high CPU for PyMuPDF OCR & PDF generation, zero egress fees to Swiss S3 storage.	**Recommended Backend** (Swiss Sovereign Stack)
AWS ECS Fargate + RDS Full-Stack Swiss Region	Serverless Containers in Zurich	Aurora PostgreSQL with automatic failover, serverless container auto-scaling during peak Swiss tax season (Jan–Mar).	**High-Volume Enterprise** (Scaling > 50k filings)
Hetzner Cloud (EU) Staging & Bootstrapped Prod	Docker Compose / Swarm	Unbeatable price-to-performance ratio (~€15–40/month), fast NVMe disks, GDPR-compliant German data centers.	**Cost Optimization** (Startups & MVP stages)

C. Optimal Production Blueprint for SunTax (Swiss Sovereign Stack)

Stack Component	Selected Provider	Configuration Specification	Est. Cost (SaaS Stage)
User Interface	Vercel Pro (Edge)	Next.js 14 SSR, custom domain app.suntax.ch, HSTS + CSP headers.	CHF ~20 / month
API & OCR Engine	Exoscale SKS (Zürich)	FastAPI 4-worker container + Celery background processor (2x vCPU, 4GB RAM).	CHF ~38 / month
Relational Database	Managed PostgreSQL 16	Encrypted at rest, automated daily snapshots, Row-Level Security (RLS).	CHF ~45 / month
Document Vault	Infomaniak Swiss S3	S3 bucket located in Geneva/Winterthur, client-side AES-256 encryption.	CHF ~5–10 / month
Total Projected Cost	**Sovereign Swiss Stack**	**Full FADP / nDSG compliance with <100ms Swiss roundtrip latency**	**CHF ~108–120 / mo**